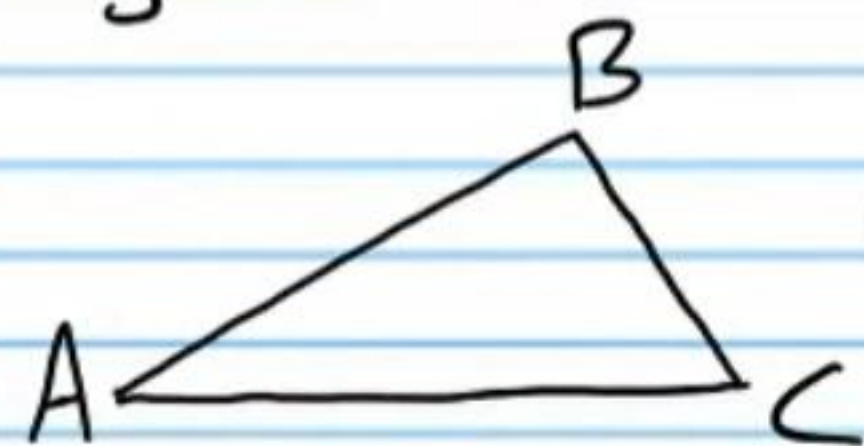


Geometry Review and the Six Basic Trigonometric Functions

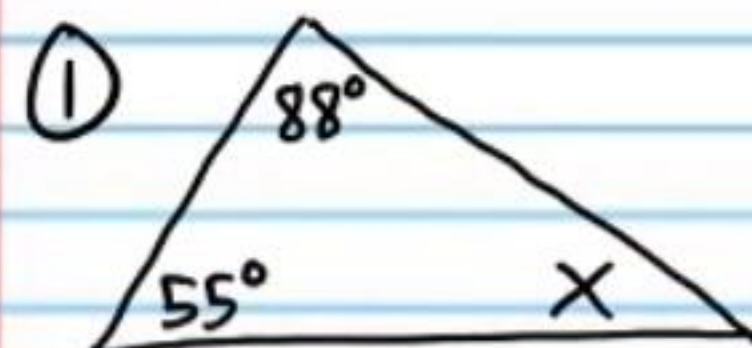
Geometry Review

The Triangle Sum Theorem: The sum of the measures of the interior angles of any triangle is 180° .



$$m\angle A + m\angle B + m\angle C = 180^\circ$$

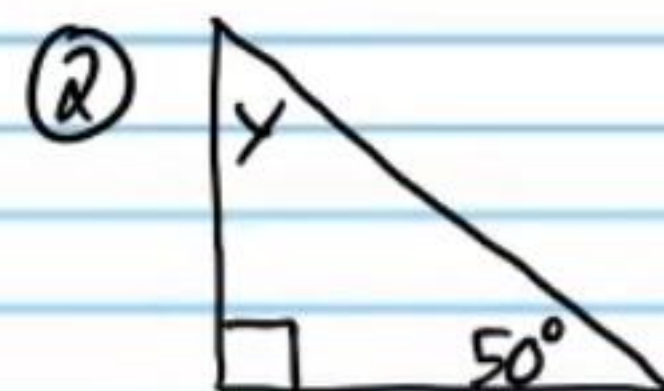
Find the value of each variable.



$$88 + 55 + x = 180$$

$$143 + x = 180$$

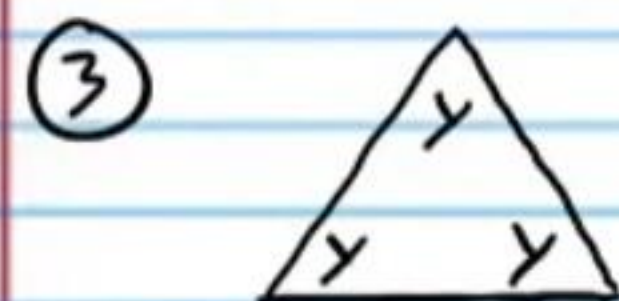
$$\boxed{x = 37^\circ}$$



$$90 + 50 + y = 180$$

$$140 + y = 180$$

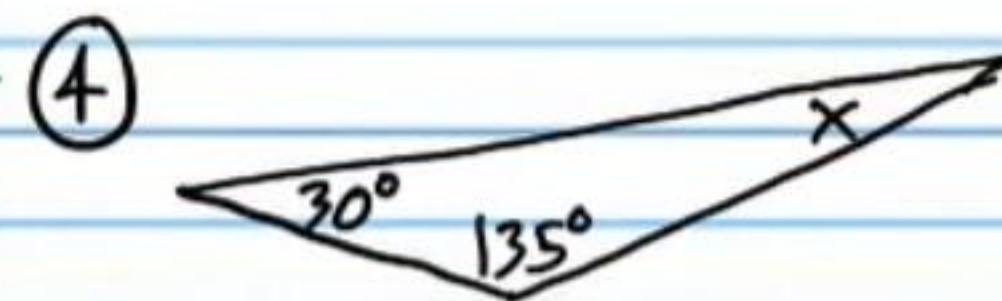
$$\boxed{y = 40^\circ}$$



$$y + y + y = 180$$

$$3y = 180$$

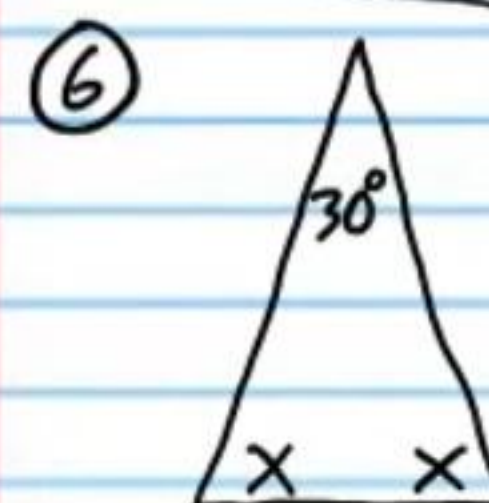
$$\boxed{y = 60^\circ}$$



$$30 + 135 + x = 180$$

$$165 + x = 180$$

$$\boxed{x = 15^\circ}$$

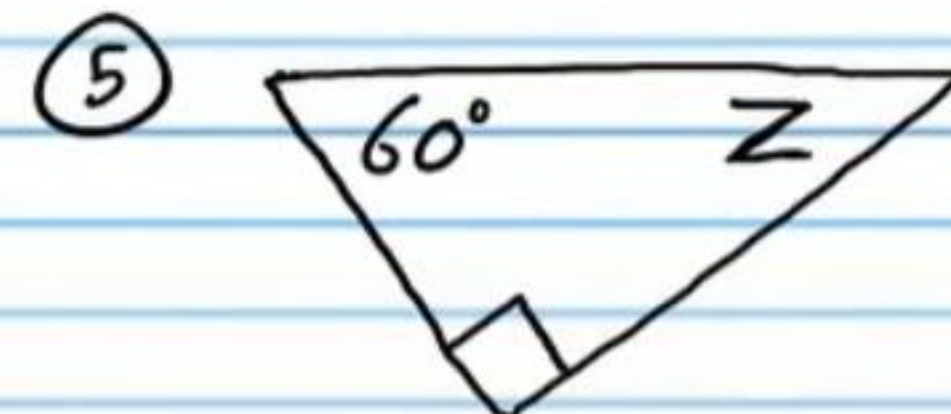


$$x + x + 30 = 180$$

$$2x + 30 = 180$$

$$2x = 150$$

$$\boxed{x = 75^\circ}$$



$$90 + 60 + z = 180$$

$$150 + z = 180$$

$$\boxed{z = 30^\circ}$$

An Introduction to Trigonometry

Pythagorean Theorem

Sides $\longleftrightarrow$ Sides

Triangle Sum Theorem

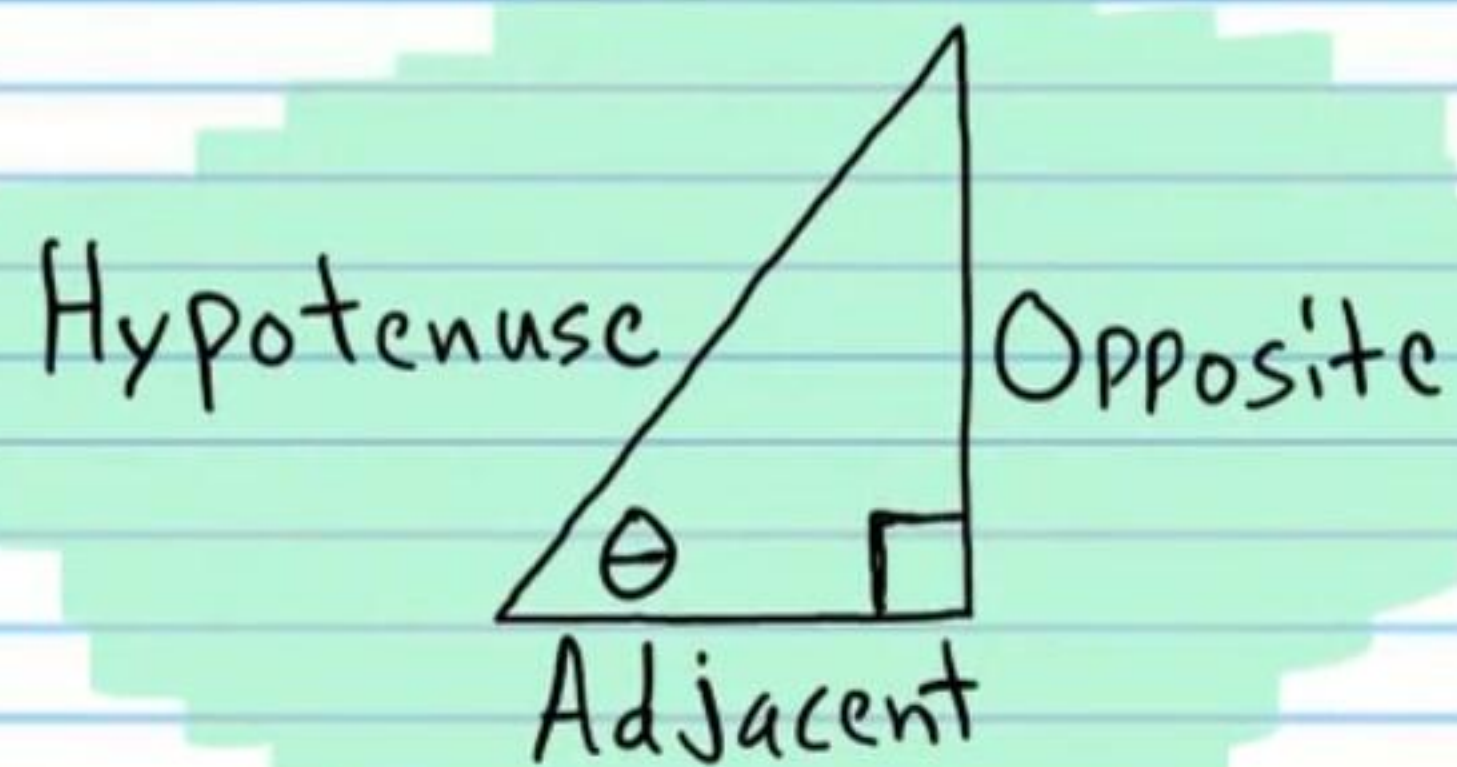
Angles $\longleftrightarrow$ Angles

Trigonometry

Sides $\longleftrightarrow$ Angles

Trigonometry: The study of relationships between sides and angles of triangles.

The Six Basic Trigonometric Functions



$$\sin \theta = \frac{\text{OPP}}{\text{HYP}}$$

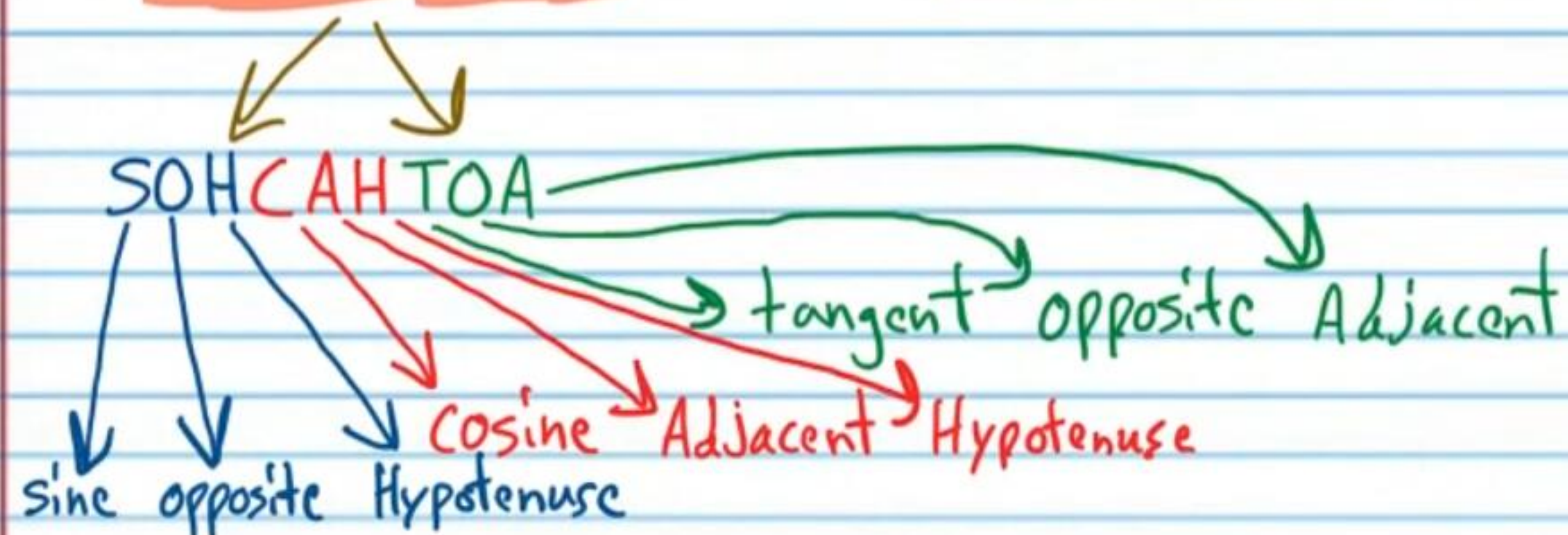
$$\csc \theta = \frac{\text{HYP}}{\text{OPP}}$$

$$\cos \theta = \frac{\text{Adj}}{\text{HYP}}$$

$$\sec \theta = \frac{\text{HYP}}{\text{Adj}}$$

$$\tan \theta = \frac{\text{OPP}}{\text{Adj}}$$

$$\cot \theta = \frac{\text{Adj}}{\text{OPP}}$$



Find the values of the expressions below.

⑦ $\sin A = \frac{15}{39} = \boxed{\frac{5}{13}}$

$\cos A = \frac{36}{39} = \boxed{\frac{12}{13}}$

$\tan A = \frac{15}{36} = \boxed{\frac{5}{12}}$

$\sin B = \frac{36}{39} = \boxed{\frac{12}{13}}$

$\cos B = \frac{15}{39} = \boxed{\frac{5}{13}}$

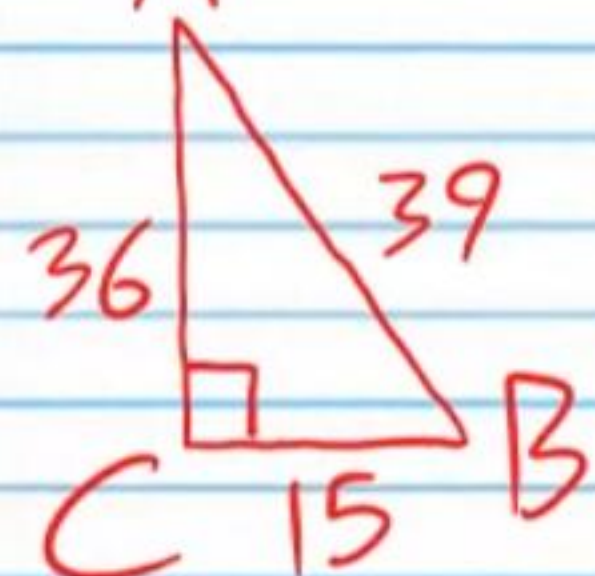
$\tan B = \frac{36}{15} = \boxed{\frac{12}{5}}$

$\csc A = \frac{39}{15} = \boxed{\frac{13}{5}}$

$\sec A = \frac{39}{36} = \boxed{\frac{13}{12}}$

$\cot A = \frac{36}{15} = \boxed{\frac{12}{5}}$

$$A \quad \csc B = \frac{39}{12} = \boxed{\frac{13}{4}}$$

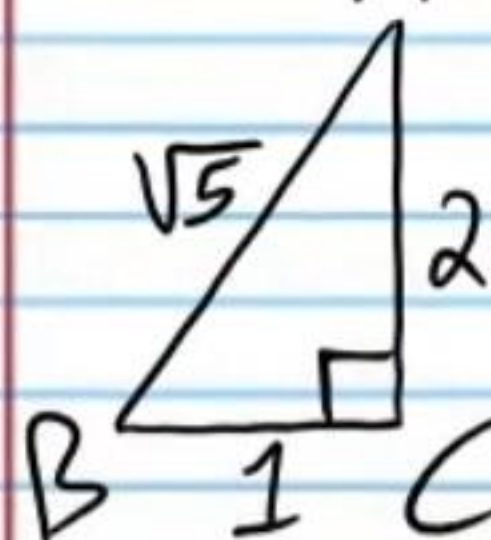


$$\sec B = \frac{39}{15} = \boxed{\frac{13}{5}}$$

$$\cot B = \frac{15}{36} = \boxed{\frac{5}{12}}$$

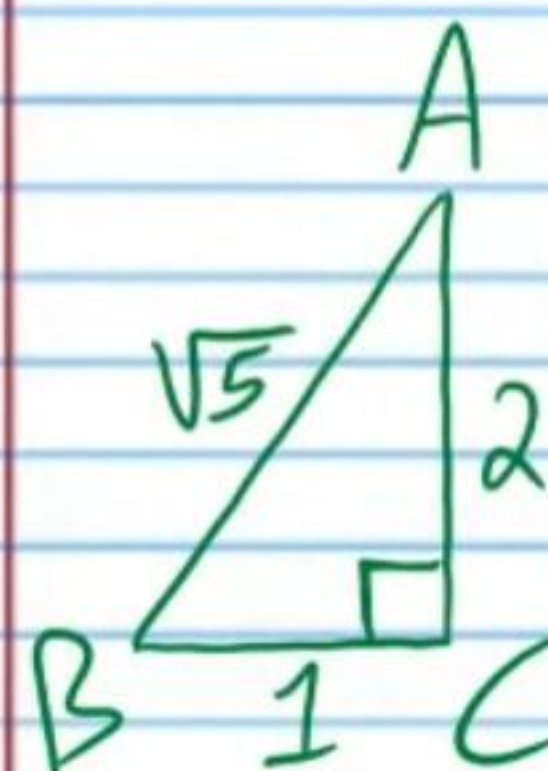
$$\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$$

$$\textcircled{8} \quad A \quad \sin A = \frac{1}{\sqrt{5}} = \frac{1 \cdot \sqrt{5}}{\sqrt{5} \cdot \sqrt{5}} = \frac{\sqrt{5}}{\sqrt{25}} = \boxed{\frac{\sqrt{5}}{5}}$$



$$\cos A = \frac{2}{\sqrt{5}} = \frac{2 \cdot \sqrt{5}}{\sqrt{5} \cdot \sqrt{5}} = \frac{2\sqrt{5}}{\sqrt{25}} = \boxed{\frac{2\sqrt{5}}{5}}$$

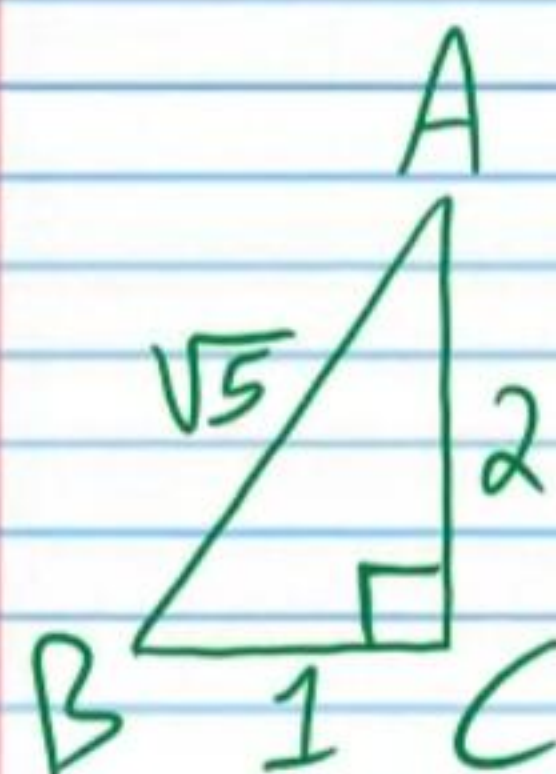
$$\tan A = \boxed{\frac{1}{2}}$$



$$\sin B = \frac{2}{\sqrt{5}} = \boxed{\frac{2\sqrt{5}}{5}}$$

$$\cos B = \frac{1}{\sqrt{5}} = \boxed{\frac{\sqrt{5}}{5}}$$

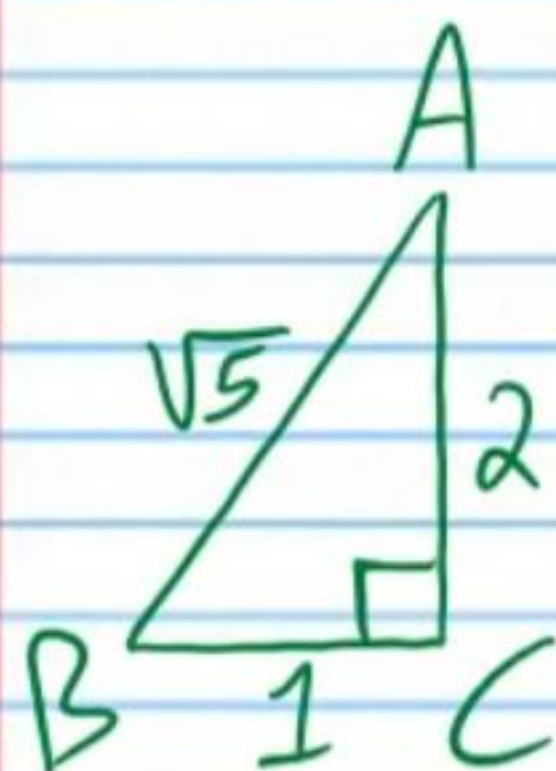
$$\tan B = \frac{2}{1} = \boxed{2}$$



$$\csc A = \frac{\sqrt{5}}{1} = \boxed{\sqrt{5}}$$

$$\sec A = \boxed{\frac{\sqrt{5}}{2}}$$

$$\cot A = \frac{2}{1} = \boxed{2}$$

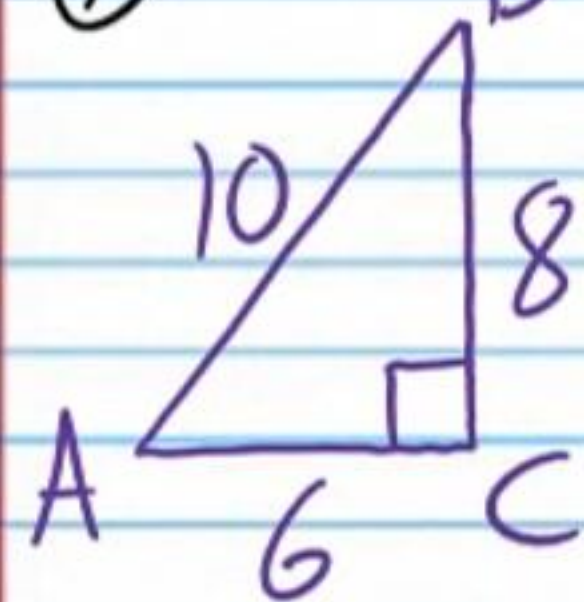


$$\csc B = \boxed{\frac{\sqrt{5}}{2}}$$

$$\sec B = \frac{\sqrt{5}}{1} = \boxed{\sqrt{5}}$$

$$\cot B = \boxed{\frac{1}{2}}$$

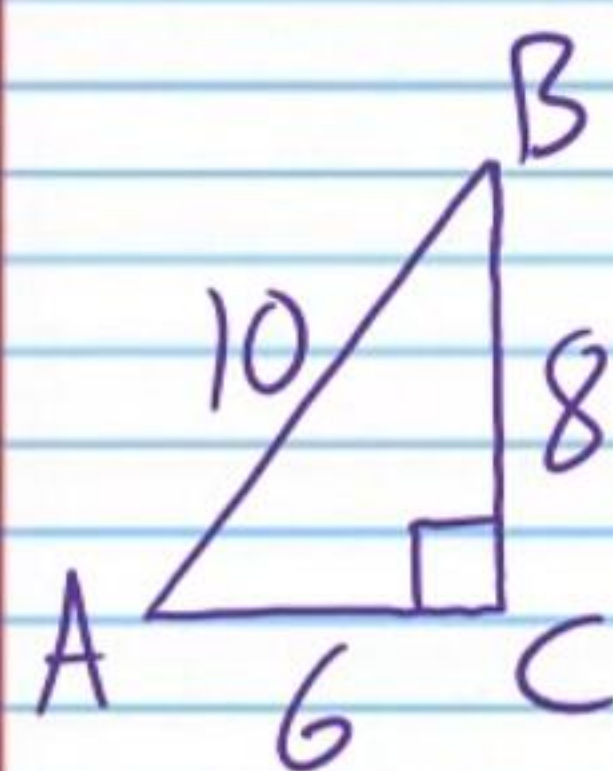
⑨ $\sin A = \frac{8}{10} = \boxed{\frac{4}{5}}$



$\cos A = \frac{6}{10} = \boxed{\frac{3}{5}}$

$\tan A = \frac{8}{6} = \boxed{\frac{4}{3}}$

$\sin B = \frac{6}{10} = \boxed{\frac{3}{5}}$



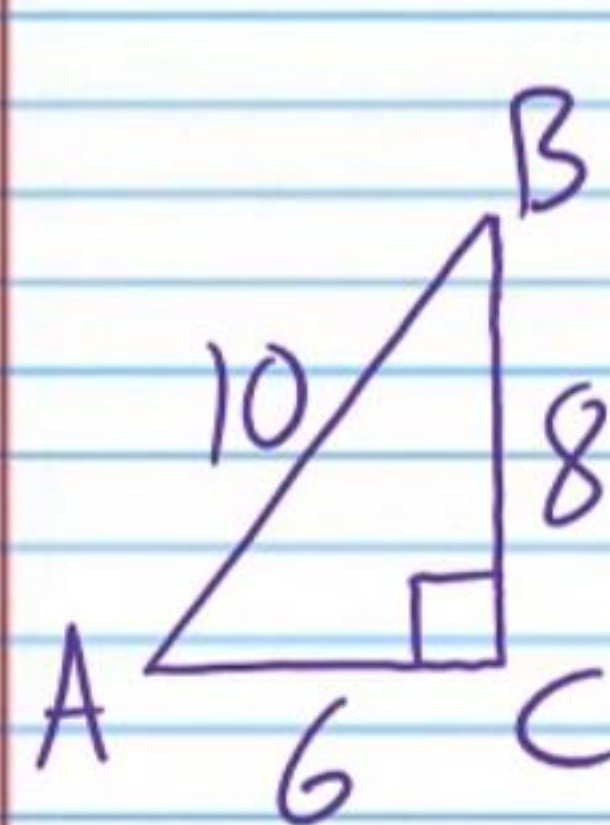
$\cos B = \frac{8}{10} = \boxed{\frac{4}{5}}$

$\tan B = \frac{6}{8} = \boxed{\frac{3}{4}}$

$\csc A = \frac{10}{8} = \boxed{\frac{5}{4}}$

$\sec A = \frac{10}{6} = \boxed{\frac{5}{3}}$

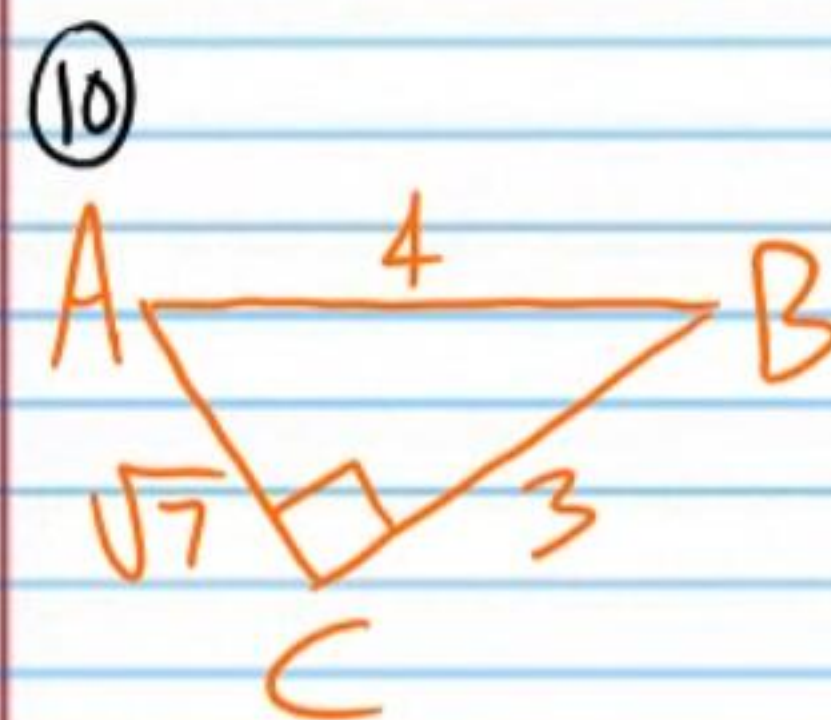
$\cot A = \frac{6}{8} = \boxed{\frac{3}{4}}$



$\csc B = \frac{10}{6} = \boxed{\frac{5}{3}}$

$\sec B = \frac{10}{8} = \boxed{\frac{5}{4}}$

$\cot B = \frac{8}{6} = \boxed{\frac{4}{3}}$

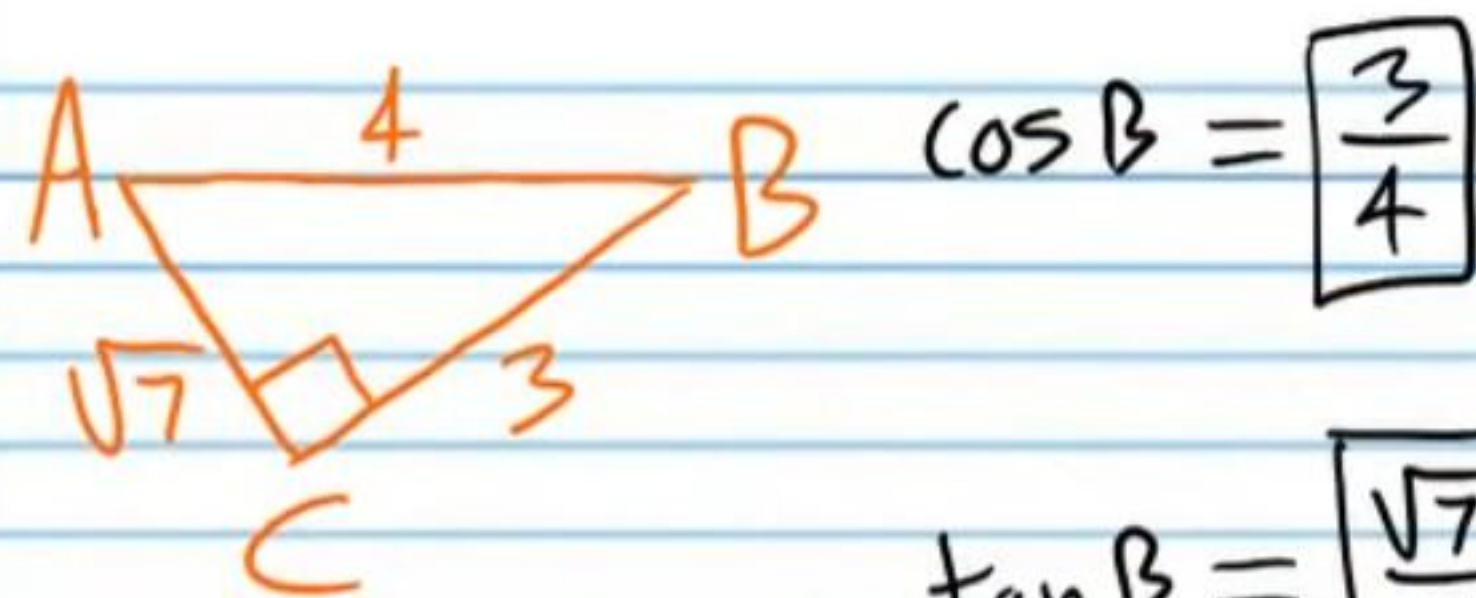


$\sin A = \boxed{\frac{3}{5}}$

$\cos A = \boxed{\frac{4}{5}}$

$\tan A = \frac{3}{4} = \frac{3 \cdot \sqrt{7}}{4 \cdot \sqrt{7}} = \frac{3\sqrt{7}}{4\sqrt{7}} = \boxed{\frac{3\sqrt{7}}{4}}$

$\sin B = \boxed{\frac{4}{5}}$



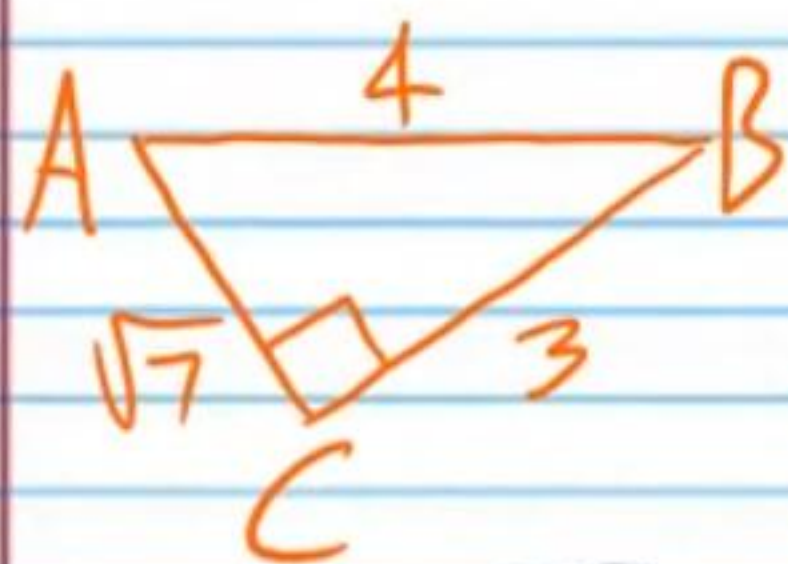
$$\cos B = \boxed{\frac{3}{4}}$$

$$\tan B = \boxed{\frac{\sqrt{7}}{3}}$$

$$\csc A = \boxed{\frac{4}{3}}$$

$$\sec A = \frac{4}{\sqrt{7}} = \frac{4 \cdot \sqrt{7}}{\sqrt{7} \cdot \sqrt{7}} = \frac{4\sqrt{7}}{7} = \boxed{\frac{4\sqrt{7}}{7}}$$

$$\cot A = \boxed{\frac{\sqrt{7}}{3}}$$



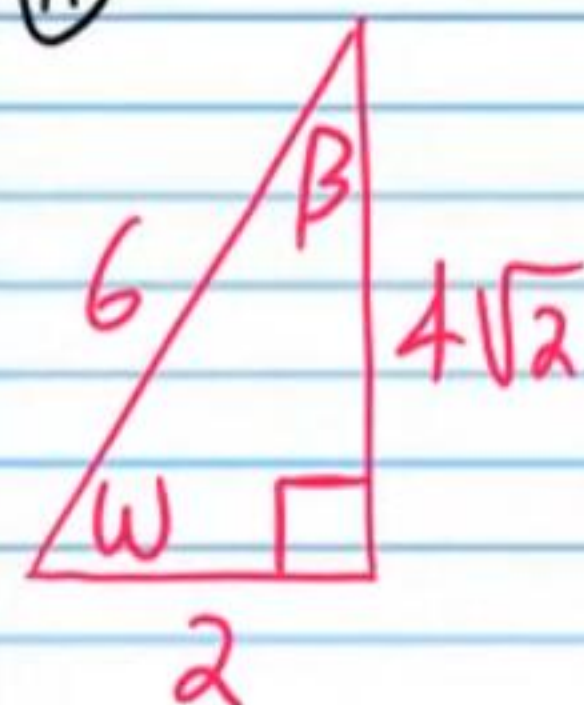
$$\csc B = \frac{4}{\sqrt{7}} = \boxed{\frac{4\sqrt{7}}{7}}$$

$$\sec B = \boxed{\frac{4}{3}}$$

$$\cot B = \frac{3}{\sqrt{7}} = \boxed{\frac{3\sqrt{7}}{7}}$$

⑪

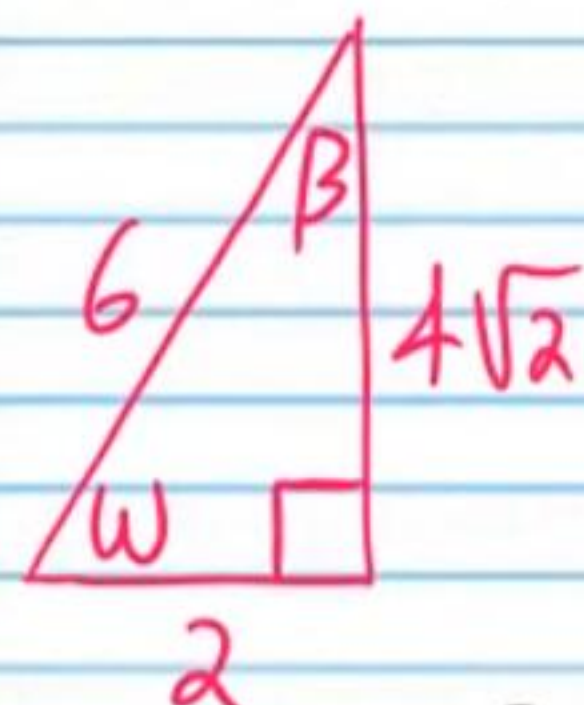
$$\sin W = \frac{4\sqrt{2}}{6} = \boxed{\frac{2\sqrt{2}}{3}}$$



$$\cos W = \frac{2}{6} = \boxed{\frac{1}{3}}$$

$$\tan W = \frac{4\sqrt{2}}{2} = \boxed{2\sqrt{2}}$$

$$\sin B = \frac{2}{6} = \boxed{\frac{1}{3}}$$

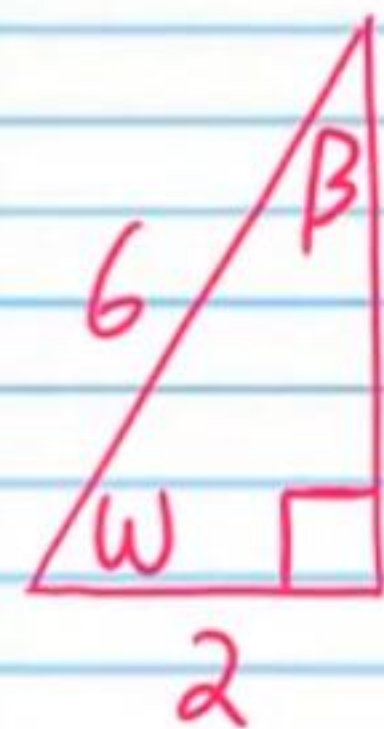


$$\cos B = \frac{4\sqrt{2}}{6} = \boxed{\frac{2\sqrt{2}}{3}}$$

$$\tan B = \frac{2}{4\sqrt{2}} = \frac{1}{2\sqrt{2}} = \frac{1 \cdot \sqrt{2}}{2\sqrt{2} \cdot \sqrt{2}} = \frac{\sqrt{2}}{2\sqrt{4}} = \frac{\sqrt{2}}{2 \cdot 2} = \boxed{\frac{\sqrt{2}}{4}}$$

$$\csc W = \frac{6}{4\sqrt{2}} = \frac{3}{2\sqrt{2}} = \frac{3 \cdot \sqrt{2}}{2\sqrt{2} \cdot \sqrt{2}} = \frac{3\sqrt{2}}{2\sqrt{4}} = \frac{3\sqrt{2}}{2 \cdot 2} = \boxed{\frac{3\sqrt{2}}{4}}$$

$$\sec W = \frac{6}{2} = \boxed{3}$$



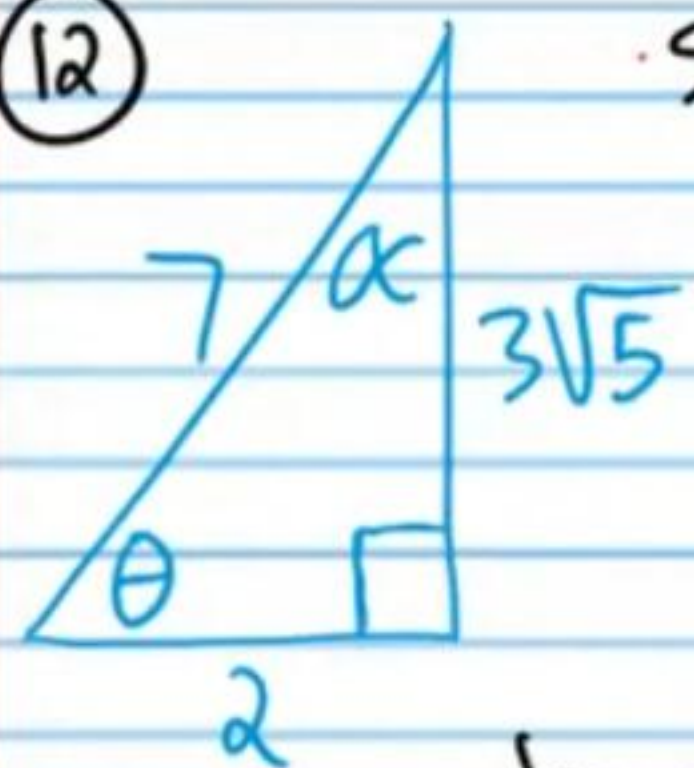
$$\cot W = \frac{2}{4\sqrt{2}} = \frac{1}{2\sqrt{2}} = \boxed{\frac{\sqrt{2}}{4}}$$

$$\csc B = \frac{6}{2} = \boxed{3}$$

$$\sec B = \frac{6}{4\sqrt{2}} = \frac{3}{2\sqrt{2}} = \boxed{\frac{3\sqrt{2}}{4}}$$

$$\cot B = \frac{4\sqrt{2}}{2} = \boxed{2\sqrt{2}}$$

(12)

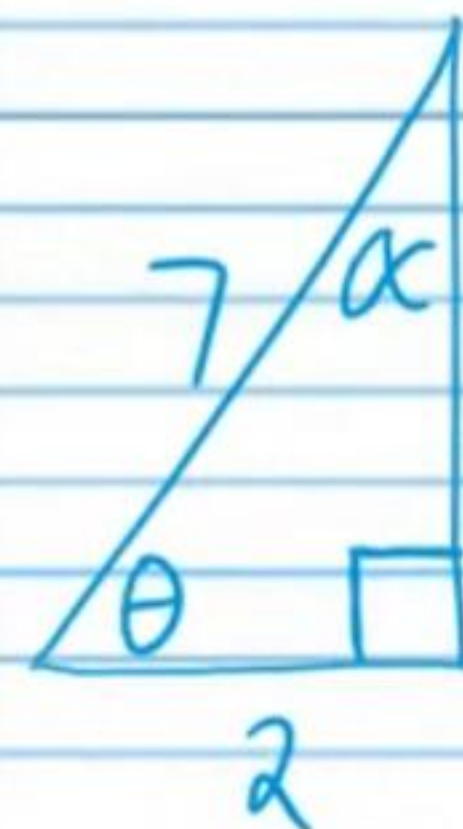


$$\sin \theta = \boxed{\frac{3\sqrt{5}}{7}}$$

$$\cos \theta = \boxed{\frac{2}{7}}$$

$$\tan \theta = \boxed{\frac{3\sqrt{5}}{2}}$$

$$\sin \alpha = \boxed{\frac{2}{7}}$$

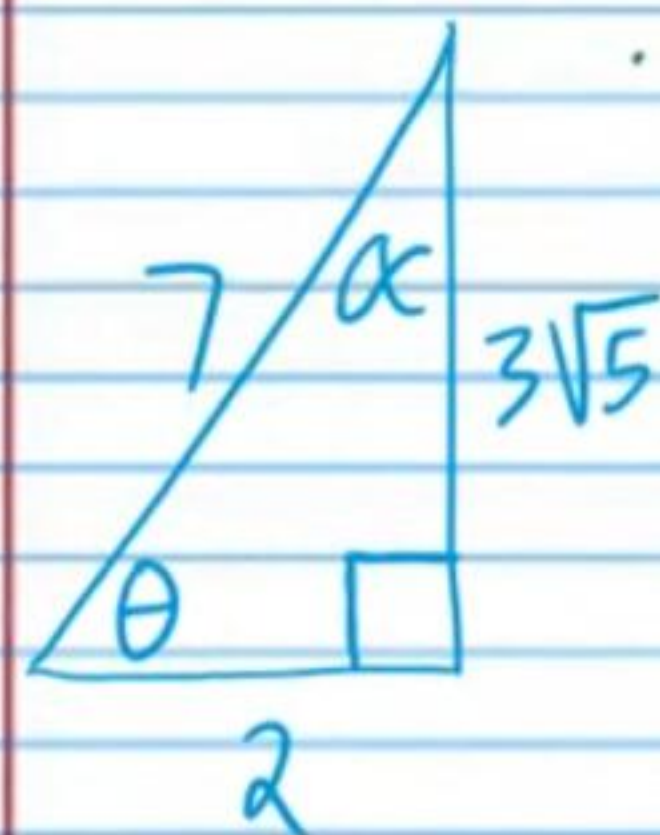


$$\cos \alpha = \boxed{\frac{3\sqrt{5}}{7}}$$

$$\cot \theta = \frac{2}{3\sqrt{5}} = \boxed{\frac{2\sqrt{5}}{15}}$$

$$\csc \theta = \frac{7}{3\sqrt{5}} = \boxed{\frac{7\sqrt{5}}{15}}$$

$$\sec \theta = \boxed{\frac{7}{2}}$$



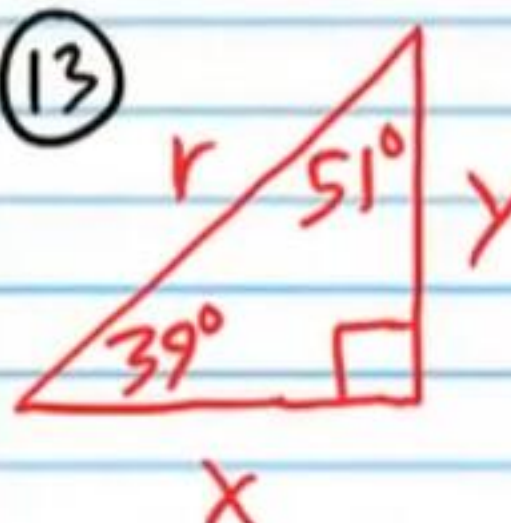
$$\tan \alpha = \frac{2}{3\sqrt{5}} = \boxed{\frac{2\sqrt{5}}{15}}$$

$$\csc \alpha = \boxed{\frac{7}{2}}$$

$$\sec \alpha = \frac{7}{3\sqrt{5}} = \boxed{\frac{7\sqrt{5}}{15}}$$

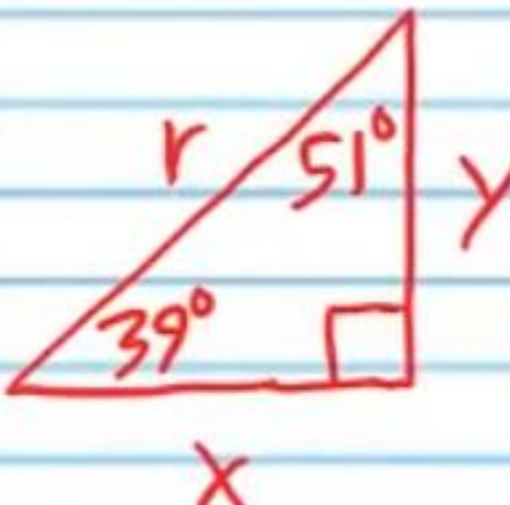
$$\cot \alpha = \boxed{\frac{3\sqrt{5}}{2}}$$

Find an exact ratio for each expression listed below.

(13)  $\sin 39^\circ = \boxed{\frac{y}{r}}$

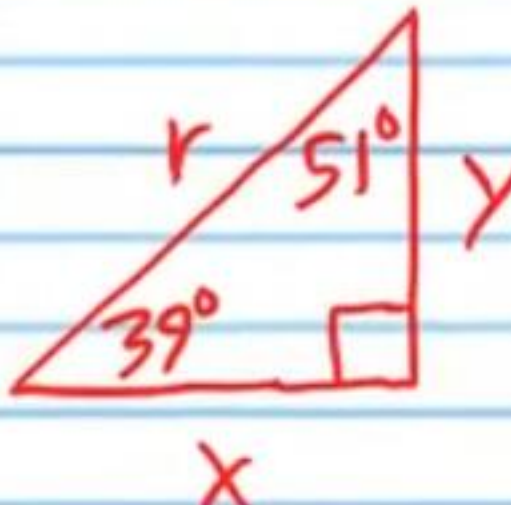
$$\cos 39^\circ = \boxed{\frac{x}{r}}$$

$$\tan 39^\circ = \boxed{\frac{y}{x}}$$

 $\sin 51^\circ = \boxed{\frac{y}{r}}$

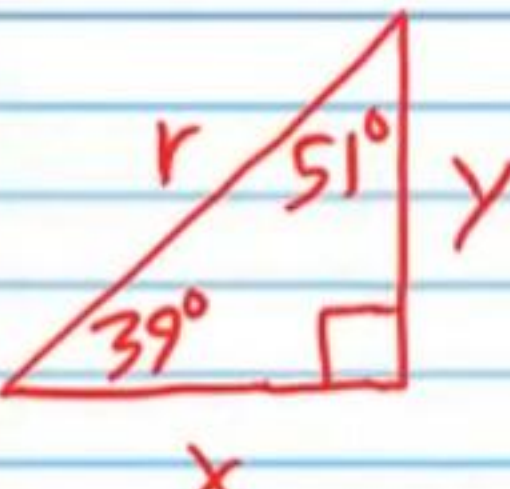
$$\cos 51^\circ = \boxed{\frac{x}{r}}$$

$$\tan 51^\circ = \boxed{\frac{y}{x}}$$

 $\csc 39^\circ = \boxed{\frac{r}{y}}$

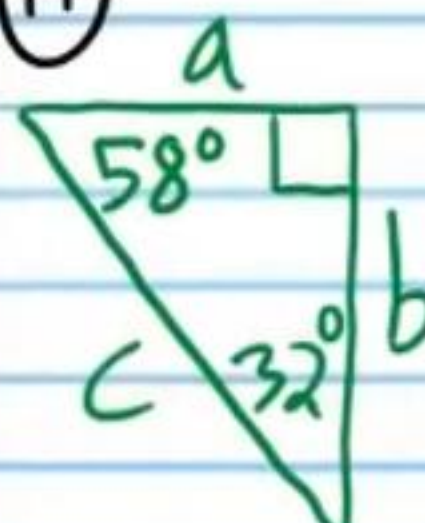
$$\sec 39^\circ = \boxed{\frac{r}{x}}$$

$$\cot 39^\circ = \boxed{\frac{x}{y}}$$

 $\csc 51^\circ = \boxed{\frac{r}{y}}$

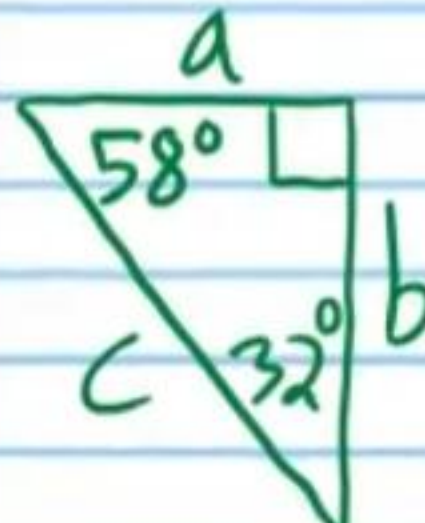
$$\sec 51^\circ = \boxed{\frac{r}{x}}$$

$$\cot 51^\circ = \boxed{\frac{x}{y}}$$

(14)  $\sin 58^\circ = \boxed{\frac{b}{c}}$

$$\cos 58^\circ = \boxed{\frac{a}{c}}$$

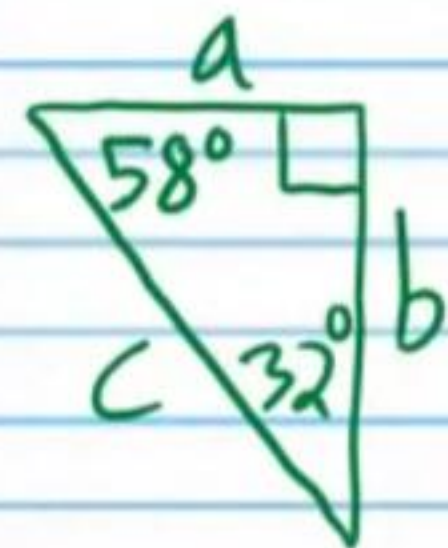
$$\tan 58^\circ = \boxed{\frac{b}{a}}$$

 $\sin 32^\circ = \boxed{\frac{a}{c}}$

$$\cos 32^\circ = \boxed{\frac{b}{c}}$$

$$\tan 32^\circ = \boxed{\frac{a}{b}}$$

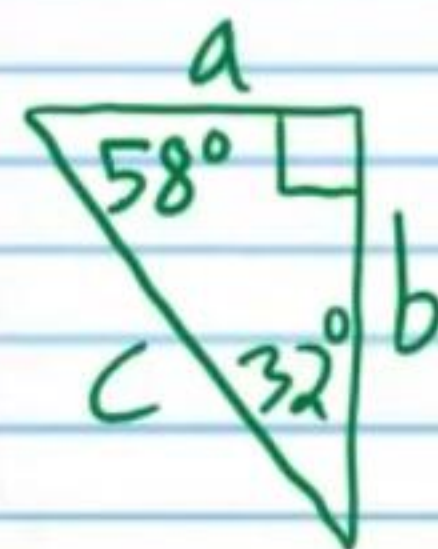
$$\cot 32^\circ = \boxed{\frac{b}{a}}$$



$$\csc 32^\circ = \boxed{\frac{c}{a}}$$

$$\sec 32^\circ = \boxed{\frac{c}{b}}$$

$$\csc 58^\circ = \boxed{\frac{c}{b}}$$



$$\sec 58^\circ = \boxed{\frac{c}{a}}$$

$$\cot 58^\circ = \boxed{\frac{a}{b}}$$

Finding Ratios with a Calculator

Use a calculator to find the sine, cosine, and tangent of each of the following angles. Round to four decimal places.

★ Be sure that your calculator is in degree mode, not radian mode.

⑮ $\theta = 82^\circ$

$$\sin 82^\circ \approx \boxed{0.9903}$$

$$\cos 82^\circ \approx \boxed{0.1392}$$

$$\tan 82^\circ \approx \boxed{7.1154}$$

⑰ $\theta = 49^\circ$

$$\sin 49^\circ \approx \boxed{0.7547}$$

$$\cos 49^\circ \approx \boxed{0.6561}$$

$$\tan 49^\circ \approx \boxed{1.1504}$$

⑯ $\theta = 25^\circ$

$$\sin 25^\circ \approx \boxed{0.4226}$$

$$\cos 25^\circ \approx \boxed{0.9063}$$

$$\tan 25^\circ \approx \boxed{0.4663}$$

⑱ $\theta = 17^\circ$

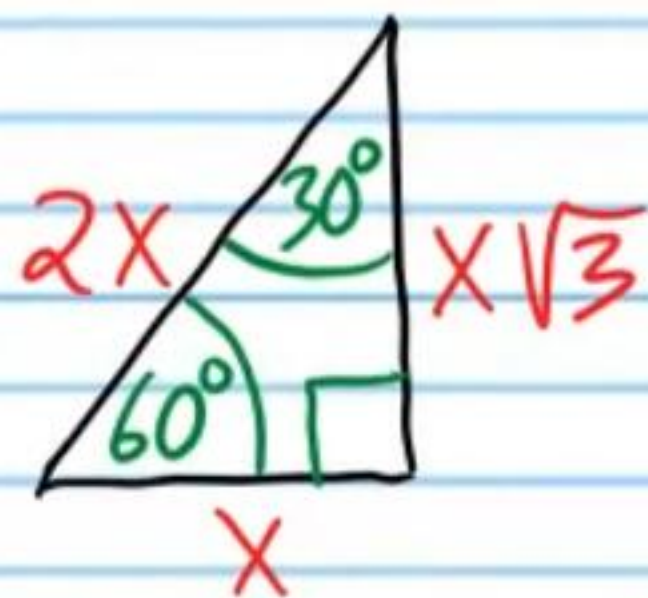
$$\sin 17^\circ \approx \boxed{0.2924}$$

$$\cos 17^\circ \approx \boxed{0.9563}$$

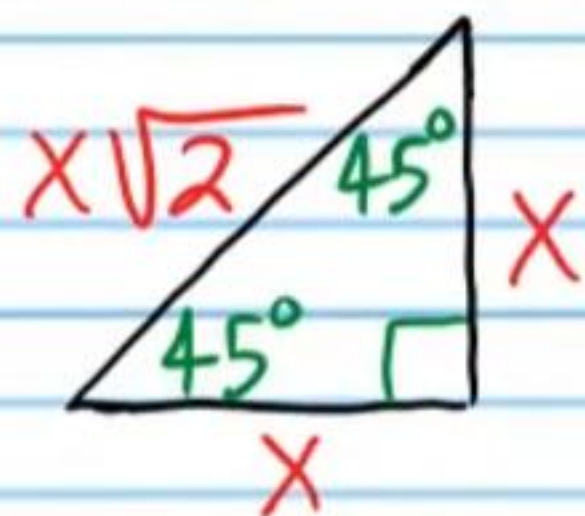
$$\tan 17^\circ \approx \boxed{0.3057}$$

Special Right Triangles

30°-60°-90° Triangle Theorem: If the inner angles of a triangle are 30°, 60°, and 90°, and the length of the side opposite the 30° angle is X , then the length of the hypotenuse is $2X$ and the length of the side opposite the 60° angle is $X\sqrt{3}$.



45°-45°-90° Triangle Theorem: If the inner angles of a triangle are 45°, 45°, and 90°, and the length of the two sides opposite the 45° angles are X , then the length of the hypotenuse is $X\sqrt{2}$.

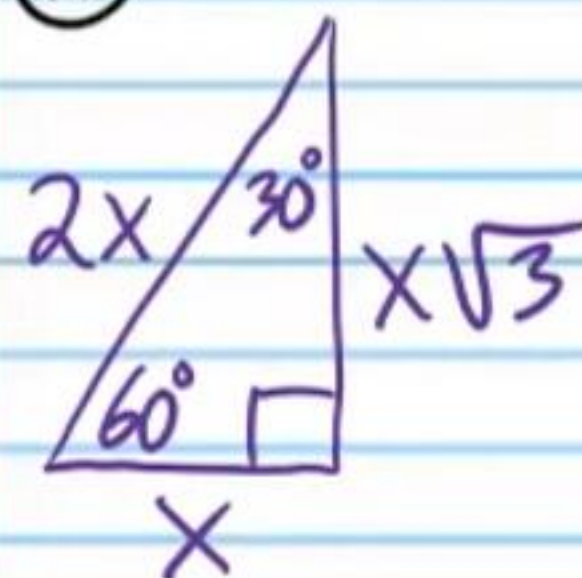


Finding Special Ratios without a Calculator

Find the value of each expression below without a calculator.

(19)

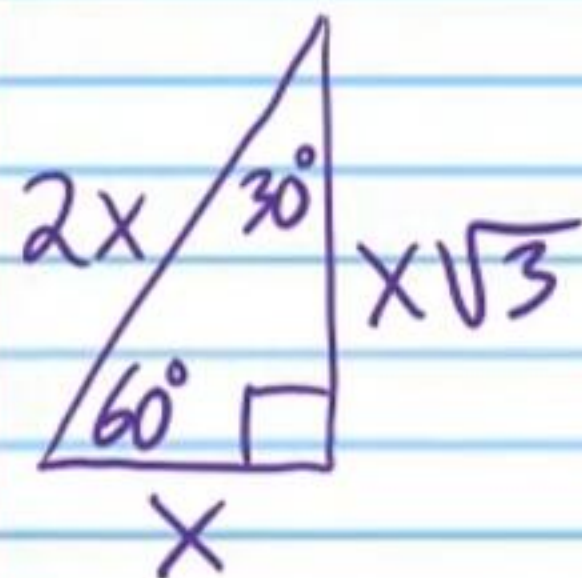
$$\sin 60^\circ = \frac{X\sqrt{3}}{2X} = \boxed{\frac{\sqrt{3}}{2}}$$



$$\cos 60^\circ = \frac{X}{2X} = \boxed{\frac{1}{2}}$$

$$\tan 30^\circ = \frac{X}{X\sqrt{3}} = \frac{1}{\sqrt{3}} = \boxed{\frac{\sqrt{3}}{3}}$$

$$\csc 30^\circ = \frac{2X}{X} = \frac{2}{1} = \boxed{2}$$



$$\sec 60^\circ = \frac{2X}{X} = \frac{2}{1} = \boxed{2}$$

$$\cot 30^\circ = \frac{X\sqrt{3}}{X} = \frac{\sqrt{3}}{1} = \boxed{\sqrt{3}}$$

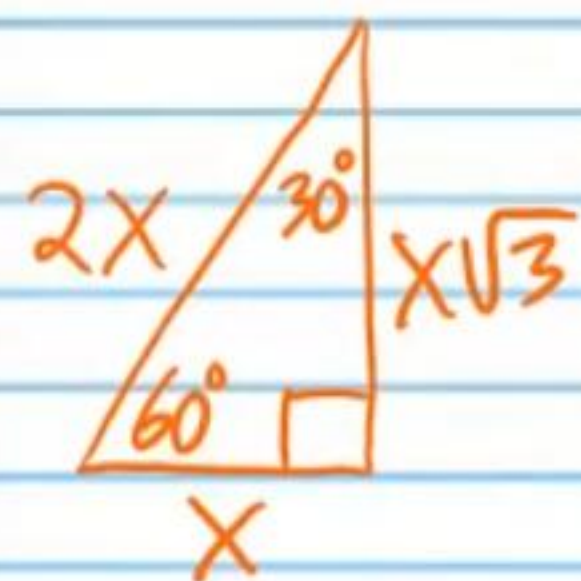
②① $\sin 30^\circ = \frac{x}{2x} = \boxed{\frac{1}{2}}$



$\cos 30^\circ = \frac{x\sqrt{3}}{2x} = \boxed{\frac{\sqrt{3}}{2}}$

$\tan 60^\circ = \frac{x\sqrt{3}}{x} = \frac{\sqrt{3}}{1} = \boxed{\sqrt{3}}$

$\csc 60^\circ = \frac{2x}{x\sqrt{3}} = \frac{2}{\sqrt{3}} = \boxed{\frac{2\sqrt{3}}{3}}$



$\sec 30^\circ = \frac{2x}{x\sqrt{3}} = \frac{2}{\sqrt{3}} = \boxed{\frac{2\sqrt{3}}{3}}$

$\cot 60^\circ = \frac{x}{x\sqrt{3}} = \frac{1}{\sqrt{3}} = \boxed{\frac{\sqrt{3}}{3}}$

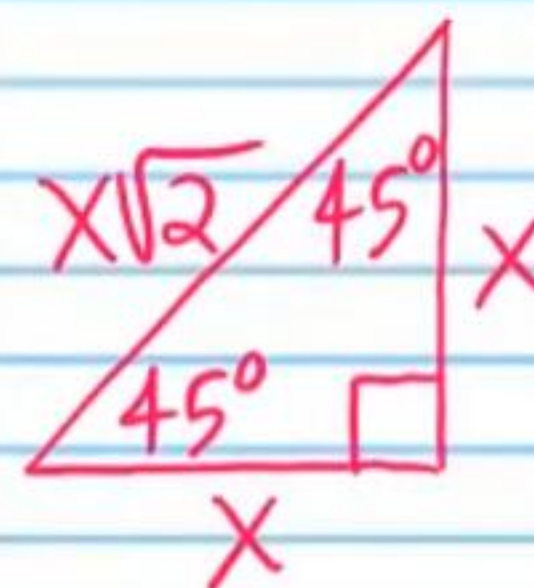
②① $\sin 45^\circ = \frac{x}{x\sqrt{2}} = \frac{1}{\sqrt{2}} = \boxed{\frac{\sqrt{2}}{2}}$



$\cos 45^\circ = \frac{x}{x\sqrt{2}} = \frac{1}{\sqrt{2}} = \boxed{\frac{\sqrt{2}}{2}}$

$\tan 45^\circ = \frac{x}{x} = \frac{1}{1} = \boxed{1}$

$\csc 45^\circ = \frac{x\sqrt{2}}{x} = \frac{\sqrt{2}}{1} = \boxed{\sqrt{2}}$



$\sec 45^\circ = \frac{x\sqrt{2}}{x} = \frac{\sqrt{2}}{1} = \boxed{\sqrt{2}}$

$\cot 45^\circ = \frac{x}{x} = \boxed{1}$

$\sin \theta = \frac{\text{OPP}}{\text{HYP}}$

$\csc \theta = \frac{\text{HYP}}{\text{OPP}}$

$\cos \theta = \frac{\text{Adj}}{\text{HYP}}$

$\sec \theta = \frac{\text{HYP}}{\text{Adj}}$

$\tan \theta = \frac{\text{OPP}}{\text{Adj}}$

$\cot \theta = \frac{\text{Adj}}{\text{OPP}}$